

Supplementary appendix

Post-transplant cyclophosphamide as graft-versus-host disease prophylaxis
after allogeneic haematopoietic stem-cell transplantation: a systematic
review and Bayesian meta-analysis of infection risk

Marta Stanzani Dimitrios P Kontoyiannis Russell E Lewis

This appendix forms part of the article and has not been edited by the journal. It is
referenced in the main text as “the appendix”.

i Scope note. The article reports the cohort-deduplicated analysis, in which one publication per cohort contributes to each outcome. Sections S1–S6 and S9–S13 apply to that analysis. The convergence diagnostics in S7 and the supplementary forest plots in S8 were produced from the pre-deduplication fits and have not yet been regenerated; where an outcome was affected by deduplication its k and pooled estimate in those two sections will not match table 2 of the article. The outcomes affected are C1 overall survival, C1 relapse-related mortality, C1 chronic GVHD (moderate to severe), C1 infection-related mortality, C2 overall survival, C2 acute GVHD, and C2 CMV reactivation.

S1. Search strategy

Table 1: Search parameters.

Field	Content
Databases searched	PubMed/MEDLINE, EMBASE (Elsevier), SCOPUS
Search dates	Database inception to [FINAL SEARCH DATE — to be taken from the Covidence/RIS export log]
Records retrieved	PubMed 1061; EMBASE 2603; SCOPUS 481 (total 4145)
Date limits applied	[to be confirmed — sources conflict]
Language limit	English
Publication types excluded via filter	None. Conference abstracts, editorials, letters, and case reports were excluded at screening rather than by search filter, to minimise inappropriate exclusion
Other sources searched	Reference list screening of included studies and prior systematic reviews; forward citation tracking (Web of Science) for key index publications; PROSPERO search for related and ongoing reviews
Deduplication and management	RIS export from each database, imported into Covidence for deduplication and screening

Field	Content
Search piloted against	Eight known included publications (Holtan 2022, Moiseev 2016, Tang 2023, Papanicolaou 2023, Salas 2023, Massoud 2021, Meyer 2024, Shaffer 2024) to verify recall

10 The protocol specifies three concept blocks combined with AND, each block an OR of
11 controlled vocabulary terms and free-text synonyms.

12 PubMed/MEDLINE

13 #1 Allogeneic HCT population

14 ("Hematopoietic Stem Cell Transplantation"[Mesh] OR "Bone Marrow Transplantation"
15 OR "allogeneic stem cell transplant*" [tiab] OR "allo-HCT" [tiab] OR "allo-
16 HSCT" [tiab]
17 OR "haploidentical" [tiab] OR "HLA-matched" [tiab] OR "matched unrelated donor" [tiab]
18 OR "mismatched unrelated donor" [tiab])

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20 #2 PTCy intervention

21 ("post-transplant cyclophosphamide" [tiab] OR "post-transplantation cyclophosphami
22 OR "posttransplant cyclophosphamide" [tiab] OR "posttransplantation cyclophospham
23 OR "PTCy" [tiab] OR "PT-Cy" [tiab] OR "PTCY" [tiab]
24 OR ("Cyclophosphamide" [Mesh] AND "Graft vs Host Disease" [Mesh]))

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26 #3 Infection outcomes

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("Infections"[Mesh] OR "Bacterial Infections"[Mesh] OR "Virus Diseases"[Mesh]
OR "Mycoses"[Mesh] OR "Bacteremia"[Mesh] OR "Sepsis"[Mesh]
OR "Cytomegalovirus Infections"[Mesh] OR "Invasive Fungal Infections"[Mesh]
OR "bacterial infection*" [tiab] OR "viral infection*" [tiab] OR "fungal infection"
OR "BSI" [tiab] OR "sepsis" [tiab] OR "CMV" [tiab] OR "EBV" [tiab] OR "HHV-
6" [tiab]
OR "BK virus" [tiab] OR "aspergillosis" [tiab] OR "candida" [tiab] OR "mucormycosis"
OR "mold" [tiab] OR "yeast" [tiab] OR "mortality" [tiab])

#4 #1 AND #2 AND #3

Filters: Humans; English

EMBASE (Emtree)

#1 Allogeneic HCT population

('bone marrow transplantation'/exp OR 'stem cell transplantation'/exp
OR 'allogeneic stem cell transplant*':ti,ab OR 'allo-hct':ti,ab OR 'allo-
hsct':ti,ab
OR haploidentical:ti,ab OR 'hla-matched':ti,ab OR 'matched unrelated donor':ti,ab
OR 'mismatched unrelated donor':ti,ab)

#2 PTCy intervention

('post-transplant cyclophosphamide':ti,ab OR 'post-transplantation cyclophospham
OR 'posttransplant cyclophosphamide':ti,ab OR 'posttransplantation cyclophospham
OR ptcy:ti,ab OR 'pt-cy':ti,ab
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```
OR ('cyclophosphamide'/exp AND 'graft versus host reaction'/exp))

#3 Infection outcomes

('infection'/exp OR 'bacterial infection'/exp OR 'virus infection'/exp OR 'mycosi
OR 'bacteremia'/exp OR 'sepsis'/exp OR 'cytomegalovirus infection'/exp
OR 'invasive fungal infection'/exp
OR 'bacterial infection*':ti,ab OR 'viral infection*':ti,ab OR 'fungal infection
OR bsi:ti,ab OR sepsis:ti,ab OR cmv:ti,ab OR ebv:ti,ab OR 'hhv-
6':ti,ab
OR 'bk virus':ti,ab OR aspergillosis:ti,ab OR candida:ti,ab OR mucormycosis:ti,a
OR mold:ti,ab OR yeast:ti,ab OR mortality:ti,ab)

#4 #1 AND #2 AND #3 AND [humans]/lim AND [english]/lim

SCOPUS

TITLE-ABS-KEY(
("allogeneic stem cell transplant*" OR "allo-HCT" OR "allo-HSCT"
OR "hematopoietic stem cell transplantation" OR "bone marrow transplantation"
OR haploidentical OR "HLA-matched" OR "matched unrelated donor"
OR "mismatched unrelated donor")
AND
("post-transplant cyclophosphamide" OR "post-transplantation cyclophosphamide"
OR "posttransplant cyclophosphamide" OR "posttransplantation cyclophosphamide"
OR PTCy OR "PT-Cy" OR (cyclophosphamide AND "graft versus host disease"))
```

73 AND
74 (infection* OR "bacterial infection*" OR "viral infection*" OR "fungal infection*" OR BSI OR sepsis OR bacteremia OR CMV OR "cytomegalovirus infection*" OR EBV
75 OR "HHV-6" OR "BK virus" OR aspergillosis OR candida OR mucormycosis OR mold
76 OR yeast OR "invasive fungal infection*" OR mortality)
77
78) AND LANGUAGE(english)

S2. Studies excluded at full-text screening (n=214)

Table 2: Reasons for exclusion at full-text screening.

Exclusion reason	n
Conference abstract	116
Single-arm or no PTCy contrast	41
Review, not primary study	21
Duplicate publication	13
Systematic review or meta-analysis	11
Non-eligible population	5
Protocol or survey only	4
Other	3
Total	214

S3. Characteristics of included studies

Full study-level (table S4a, 251 rows) and arm-level (table S4b, 525 rows) characteristics are provided as supplementary data files (`Table_S4a_study_characteristics.csv`, `Table_S4b_arm_characteristics.csv`), and are reproduced from the extraction database in the accompanying notebook *Study and arm characteristics*.

S5. Risk of bias

Table 3: ROBINS-I domain-level risk of bias, counts by judgement (n = 211 comparative observational studies; single-arm descriptive studies excluded).

Domain	Low	Moderate	Serious
D1 Confounding	0	83	128
D2 Selection of participants	26	140	45
D3 Classification of interventions	51	155	5
D4 Deviations from intended interventions	193	18	0
D5 Missing data	70	140	1
D6 Measurement of outcomes	101	101	2
D7 Selection of the reported result	161	42	1

Full justification narratives are available in `rob.csv`, submitted as a supplementary data file.

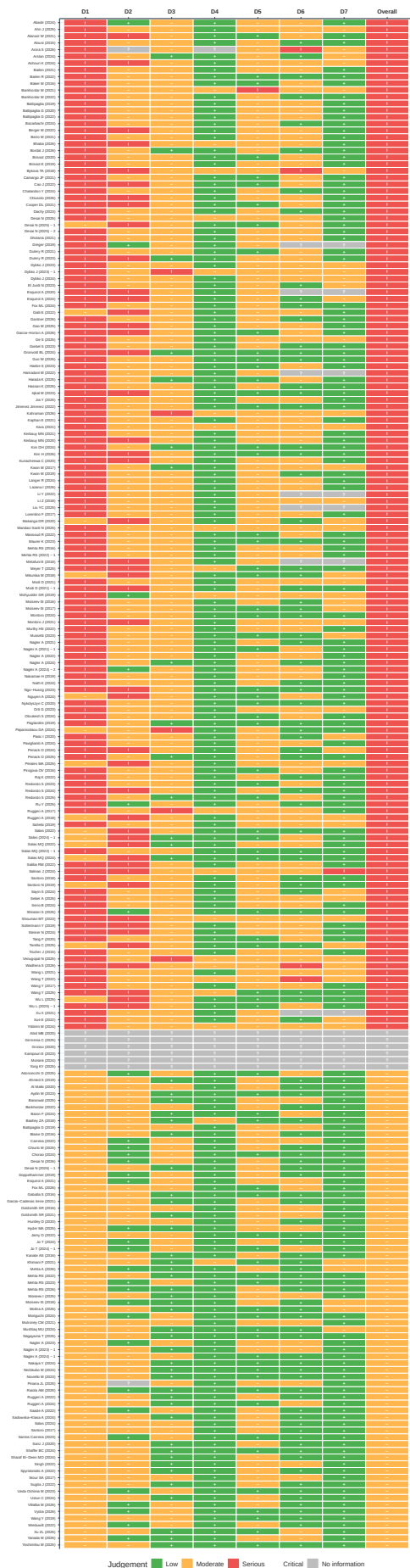


Figure 1: ROBINS-I traffic-light plot for all comparative observational studies.

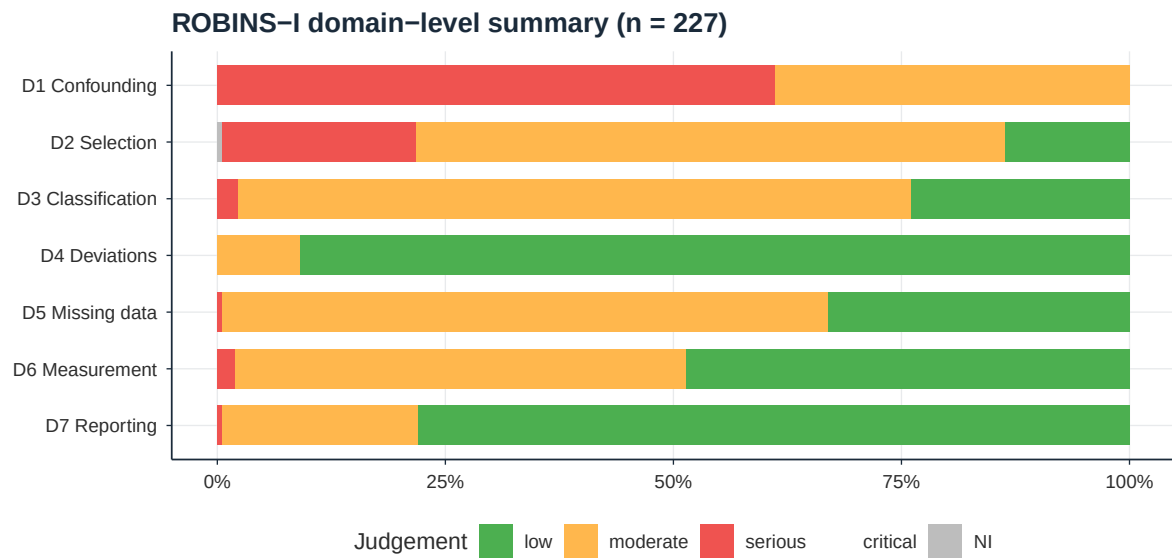


Figure 2: ROBINS-I domain-level summary.

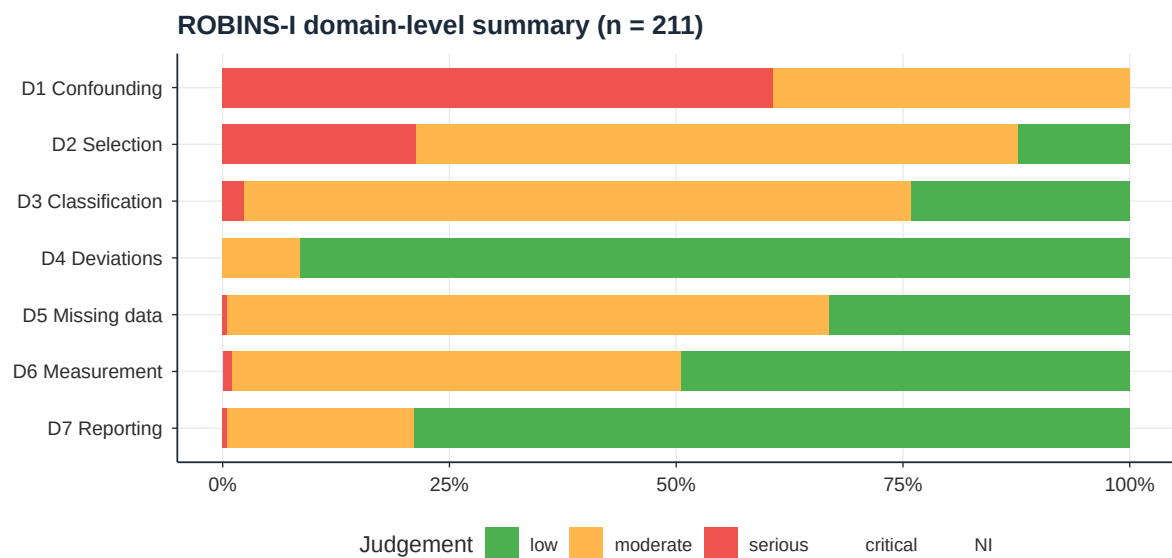


Figure 3: RoB 2 summary for the 14 randomised controlled trials.

S6. Model specification and priors

Primary model (M1), for study s , arm i :

$$\text{events}_i \mid n_i \sim \text{Binomial}\left(n_i, \text{logit}^{-1}\left(\alpha + \beta_{\text{PTCy}}x_i + \beta_{\text{tp}}t_i + u_{s[i]}\right)\right), \quad u_s \sim \mathcal{N}(0, \tau^2)$$

where x_i indicates a PTCy arm and t_i indicates a late (1 year) rather than early (day +100 or +180) timepoint.

Table 4: Prior specification for the primary model.

Parameter	Prior
Fixed effects ()	$\text{N}(0, 2 \cdot 5)$
Intercept ()	$\text{N}(0, 1 \cdot 5)$
Between-study SD ()	Student $t(3, 0, 1)$, half

The steroid-mediation model (M2) adds arm-level systemic steroid exposure percentage, mean-centred, as an additional fixed effect.

MCMC: four chains $\times$ 4000 iterations (1000 warmup), `adapt_delta` = 0.95, giving 12 000 post-warmup draws per model.

S7. MCMC diagnostics

All Bayesian models were fitted with `brms` v2.23.0 and the `rstan` backend. R (threshold $< 1 \cdot 01$), bulk and tail effective sample size (ESS; threshold > 400), and the number of divergent transitions are reported for every fitted model. Values are the worst case (maximum R , minimum ESS) across all model parameters.

The diagnostics below are from the pre-deduplication fits; see the scope note at the head of this appendix.

Table 5: MCMC convergence diagnostics, comparison 1 models.

Model	Max R	Min bulk ESS	Min tail ESS	Divergences
OS M1	$1 \cdot 0021$	1289	2891	0
OS M2	$1 \cdot 0020$	2579	4642	0
NRM M1	$1 \cdot 0017$	4176	5261	0
NRM M2	$1 \cdot 0017$	3061	4066	0
aGVHD M1	$1 \cdot 0012$	2178	3317	0
CMV M1	$1 \cdot 0011$	2814	4158	0
CMV M2	$1 \cdot 0008$	3323	4990	0
BSI M1	$1 \cdot 0011$	3643	5034	0
IFI M1	$1 \cdot 0021$	2795	4074	0
RRM M1	$1 \cdot 0006$	3872	5960	0
BK M1	$1 \cdot 0013$	3099	4143	0
IRM M1	$1 \cdot 0017$	2491	3652	0

Table 6: MCMC convergence diagnostics, comparison 2 models.

Model	Max $\mathcal{R}$	Min bulk ESS	Min tail ESS	Divergences
OS M1	$1 \cdot 0007$	3103	4585	0
OS M2	$1 \cdot 0009$	3494	5306	0
NRM M1	$1 \cdot 0035$	2876	2293	0
aGVHD M1	$1 \cdot 0015$	3260	4723	0
CMV M1	$1 \cdot 0012$	3693	5243	0
CMV M2	$1 \cdot 0008$	3384	5554	0
RRM M1	$1 \cdot 0002$	3225	4933	0
BK M1	$1 \cdot 0592$	67	17	0

103 All models except C2 BK achieved $\mathcal{R} < 1 \cdot 01$ with zero divergent transitions and bulk
104 ESS above 1000. The C2 BK model ($k = 3$) did not converge to that standard ($\mathcal{R} =$
105 $1 \cdot 0592$, bulk ESS 67, tail ESS 17) and its posterior should be read as indicative only.

106 All models show adequate calibration, with points clustered near the identity line and
107 symmetric scatter. The C1 IFI model ($k = 6$) shows the widest scatter, consistent with
108 its large posterior (figure 5) and the heterogeneity identified in S9. No systematic over-
109 or under-prediction is evident in any model.

Figure S7b: Trace Plots — OS M1 and CMV M1

Treatment effect (b_{ptcy_binary} , log-OR) and between-study heterogeneity SD (τ) for the two most informative Comparison 1 models. 4 chains $\times$ 3,000 post-warmup iterations.

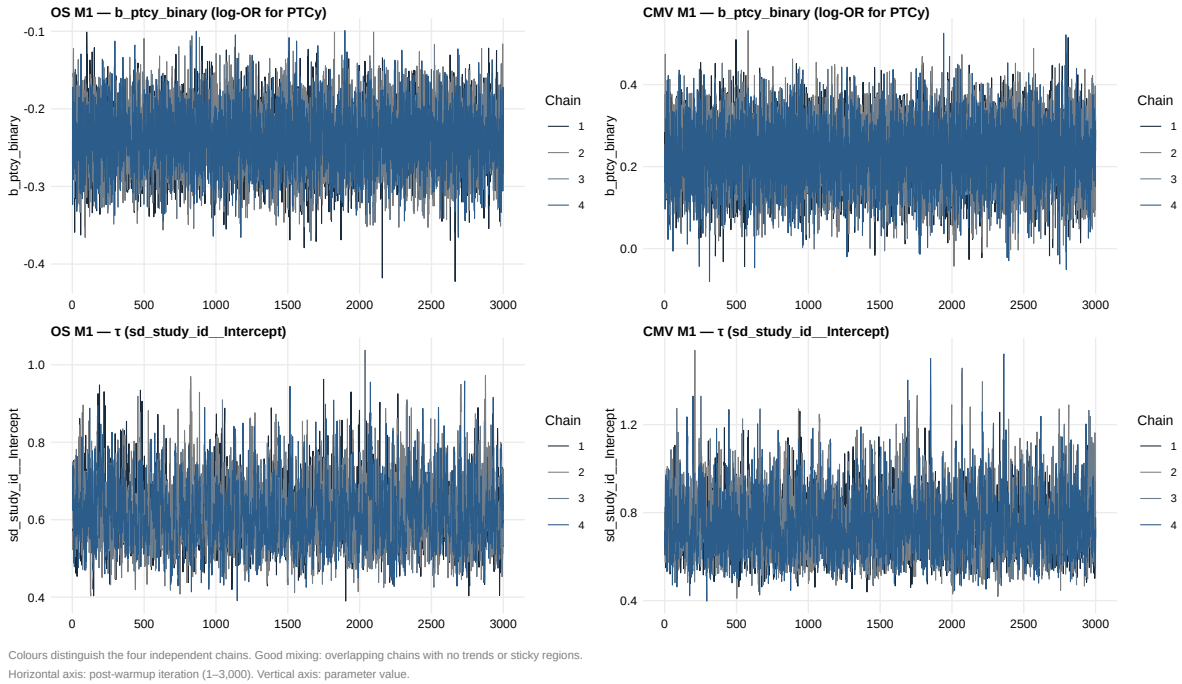


Figure 4: Trace plots for b_{ptcy_binary} (top) and $sd_study_id_Intercept$ (, bottom) in OS M1 (left) and CMV M1 (right). All four chains mix well across the full post-warmup phase.

S8. Forest plots

Individual study odds ratios with 95% CIs and the pooled frequentist random-effects estimate (`metafor`, REML), shown alongside the Bayesian estimates reported in the main text as concordance checks.

These plots are from the pre-deduplication fits; see the scope note at the head of this appendix. For the deduplicated version of the two forest plots that changed, see the *Forest plots* notebook accompanying the article.

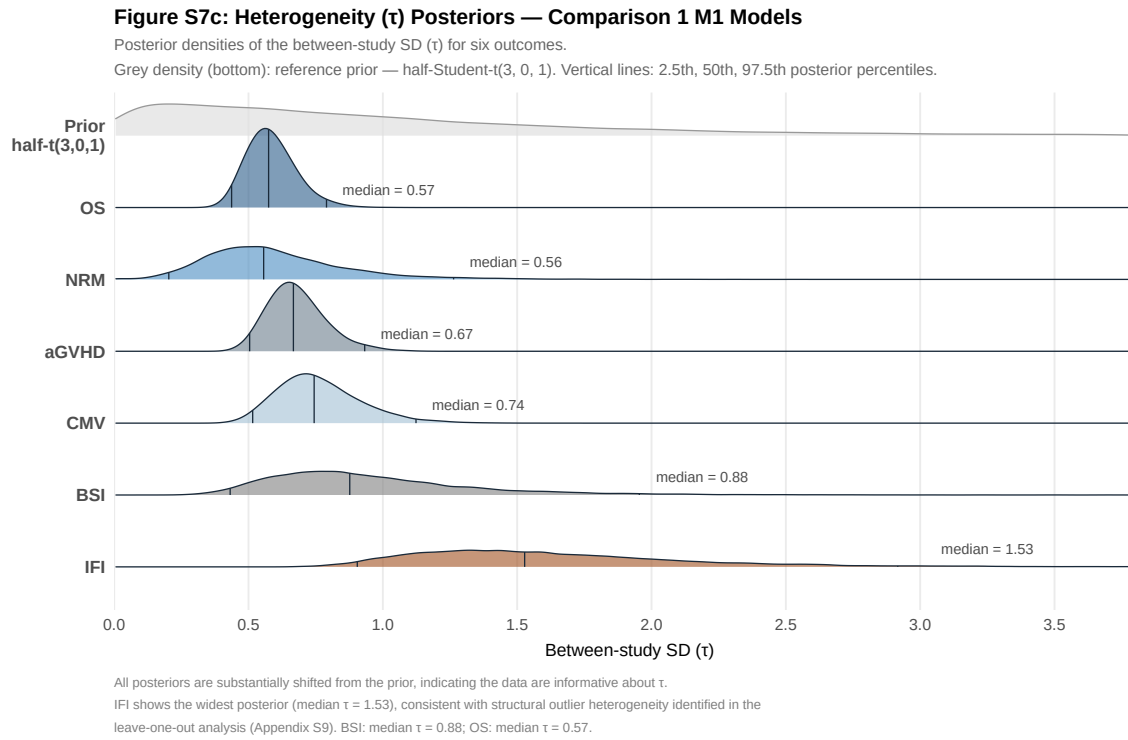


Figure 5: Posterior densities of (between-study SD) for comparison 1 M1 models. The half-Student $t(3, 0, 1)$ prior common to all models is shown in grey. IFI shows the widest posterior, consistent with the structural heterogeneity identified in the leave-one-out analysis (S9).

Figure S7d-i: Posterior Predictive Checks — Comparison 1 (PTCy vs CNI+MTX)

Each point = one study arm. Observed event proportion (x) vs. posterior predictive mean proportion (y).

Red dashed line = identity (perfect calibration). Points should scatter symmetrically around the line.

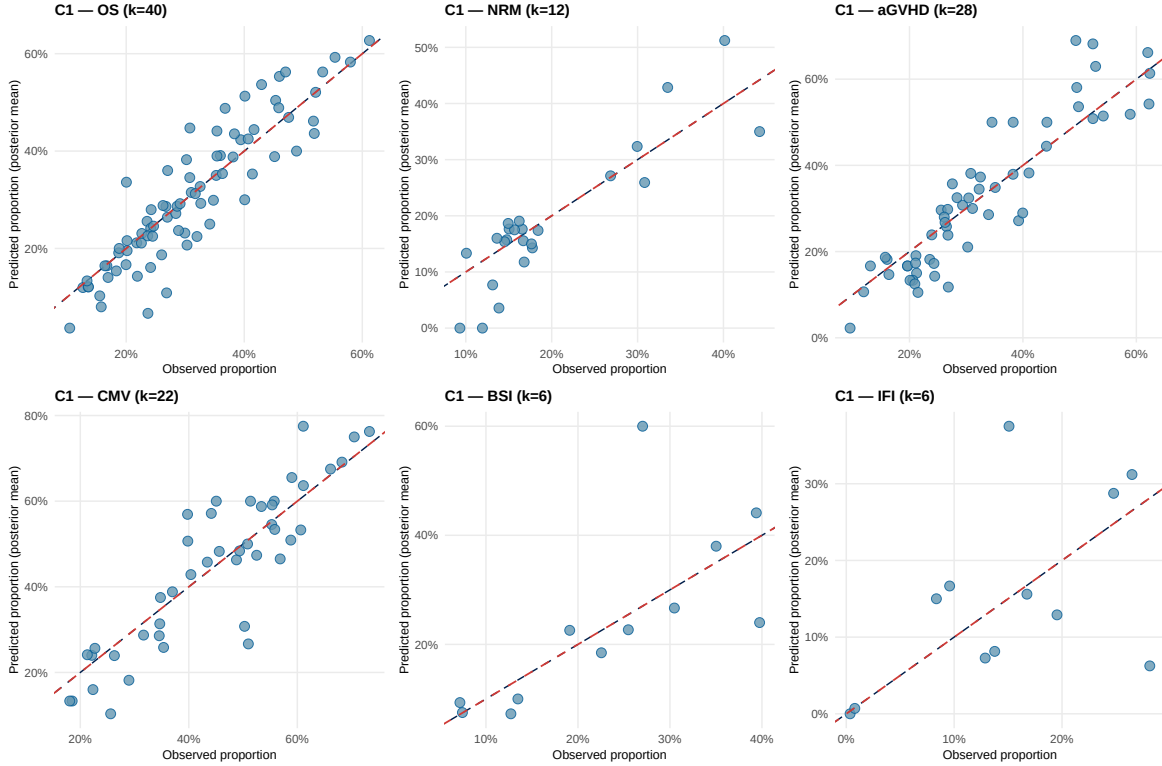


Figure 6: Posterior predictive checks, comparison 1. Each point is one study arm: the observed event proportion against its posterior predictive mean, over 500 posterior draws. The dashed line is the identity.

Figure S7d-ii: Posterior Predictive Checks — Comparison 2 (PTCy vs ATG)

Each point = one study arm. Observed event proportion (x) vs. posterior predictive mean proportion (y).

Red dashed line = identity (perfect calibration).

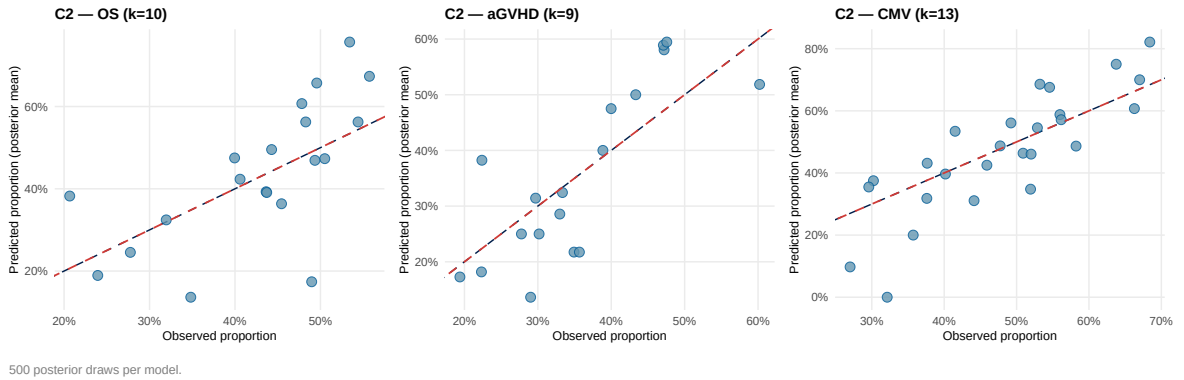
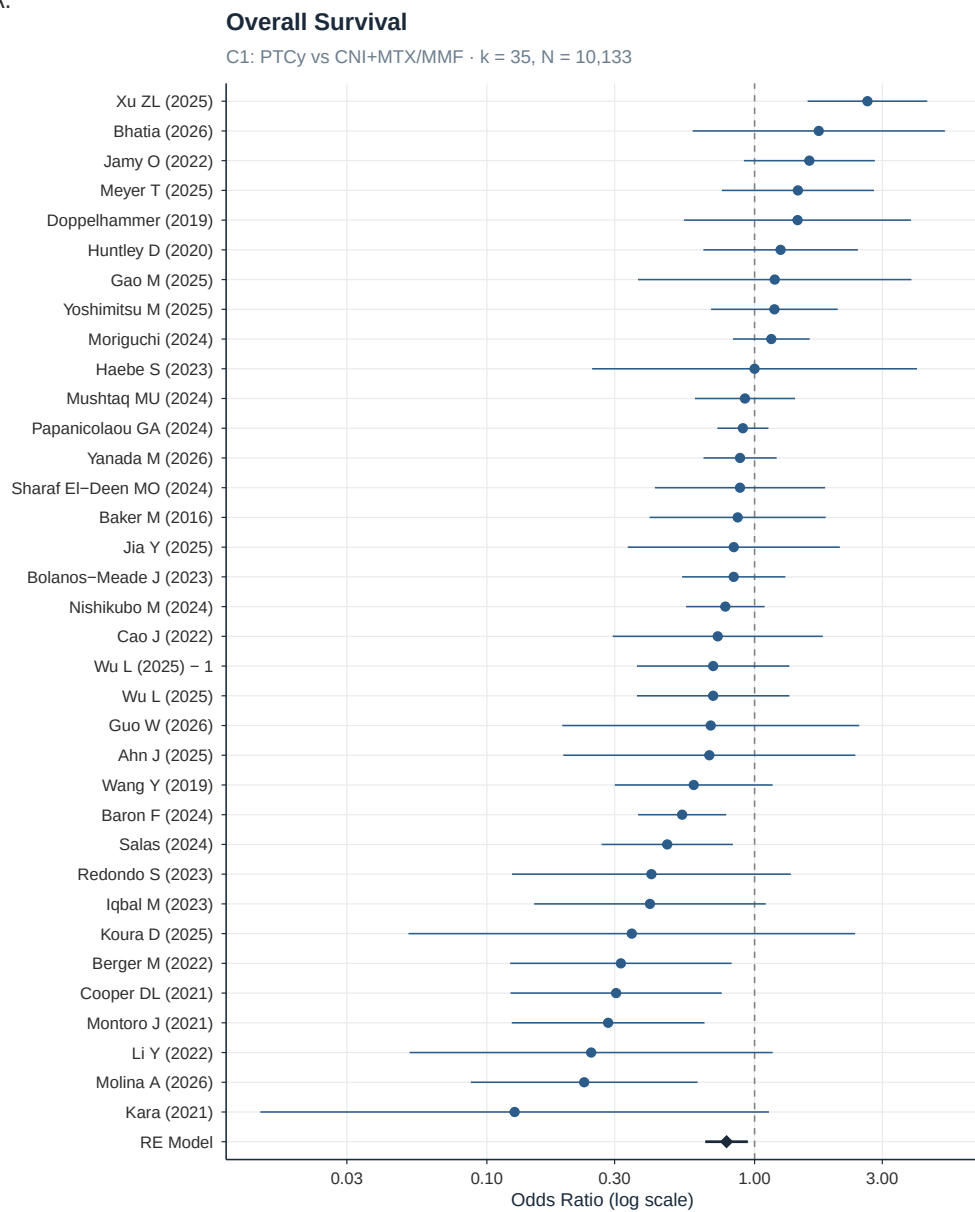


Figure 7: Posterior predictive checks, comparison 2. Layout and interpretation as in figure 6.

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B.

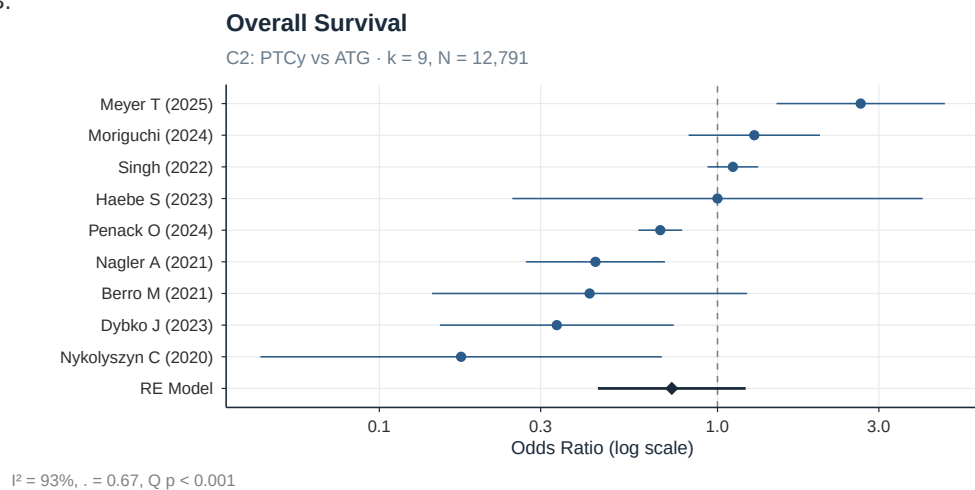
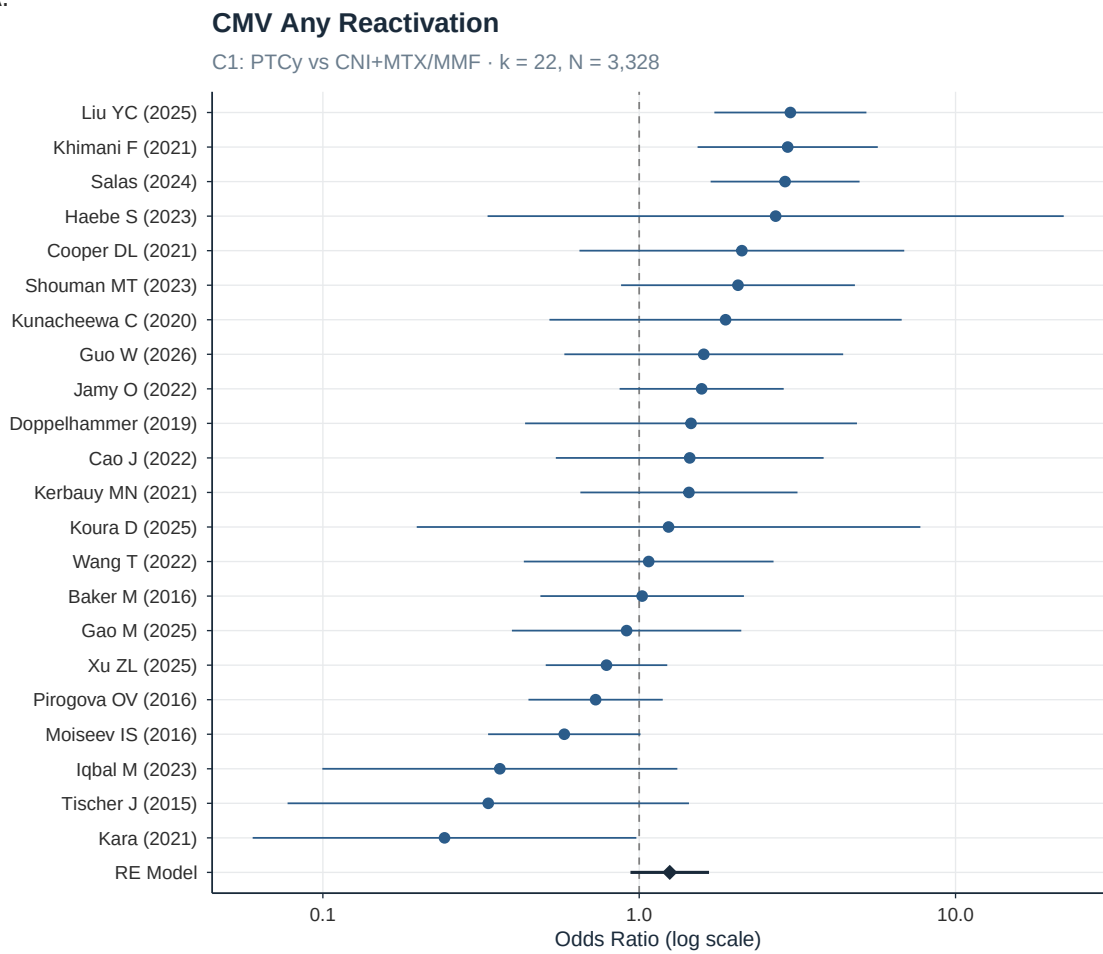


Figure 8: Overall survival, frequentist REML concordance check for figure 2 of the main text.

A.



B.

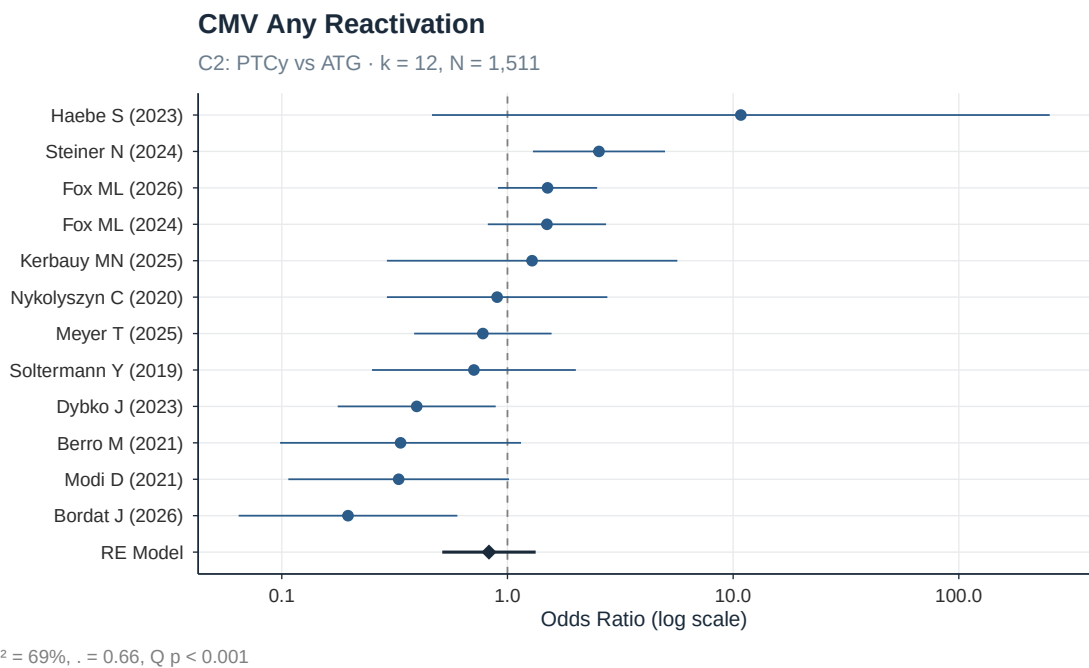
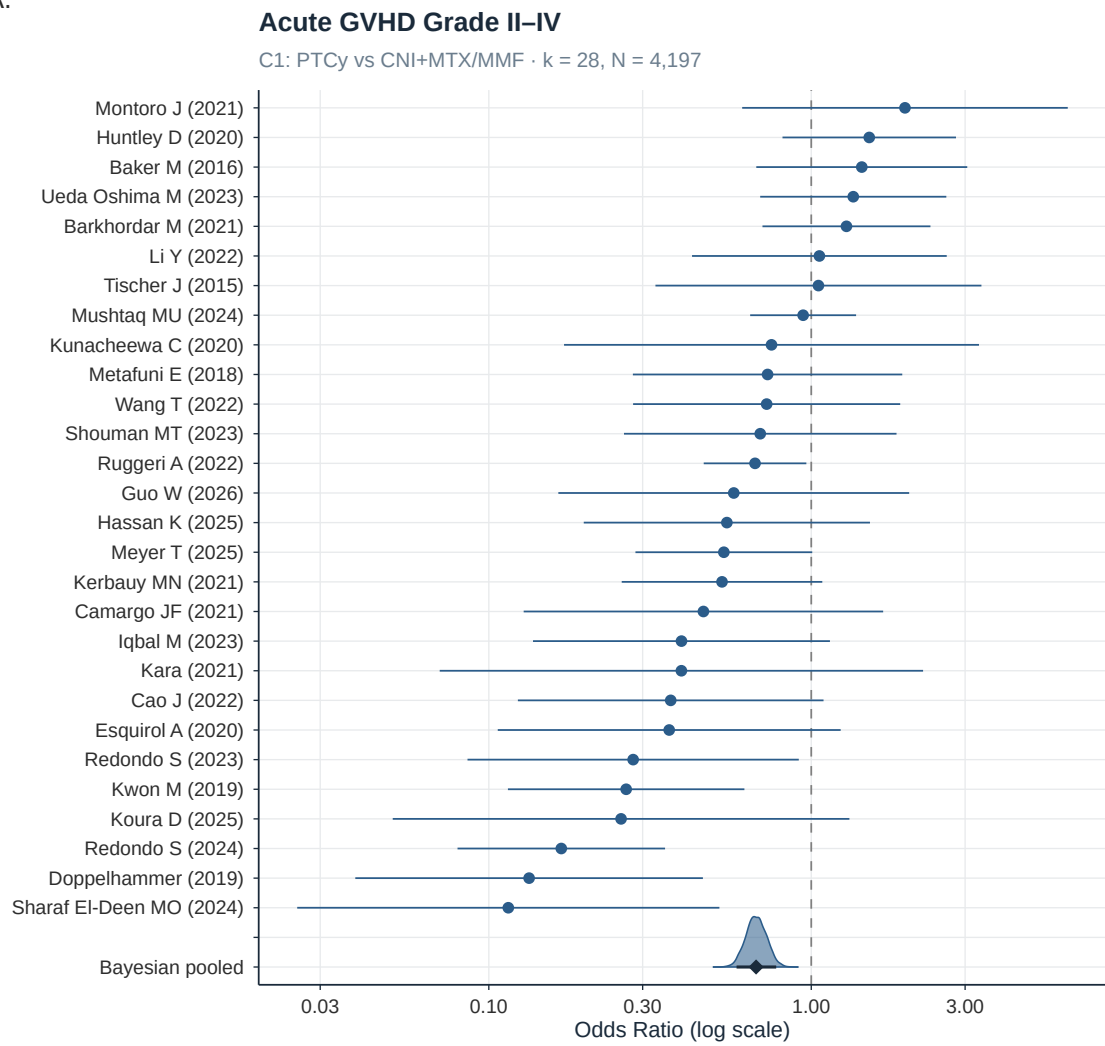


Figure 9: CMV any-reactivation, frequentist REML concordance check for figure 3 of the main text.

A.



B.

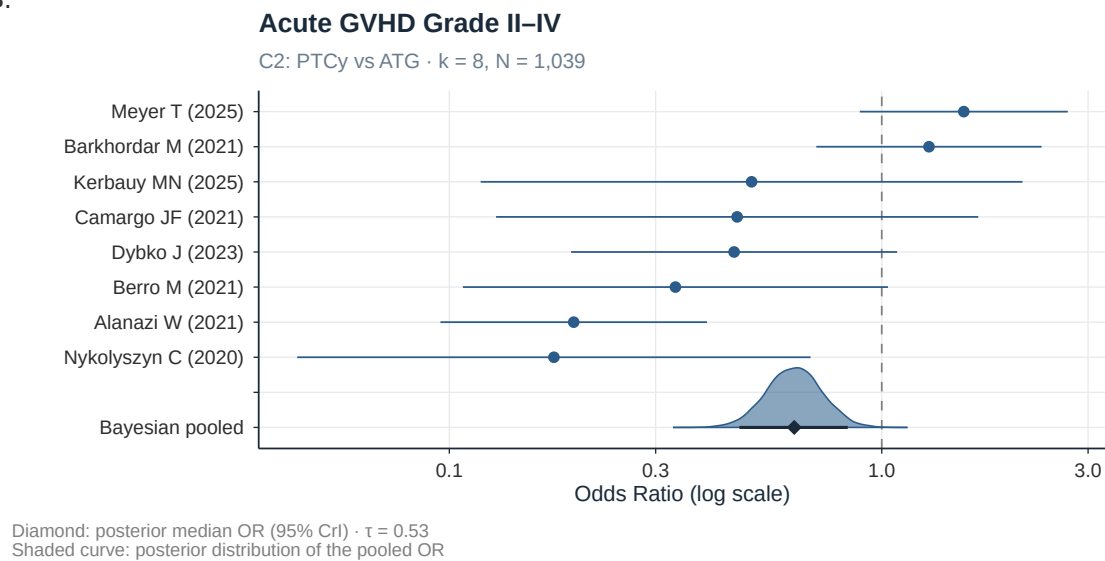
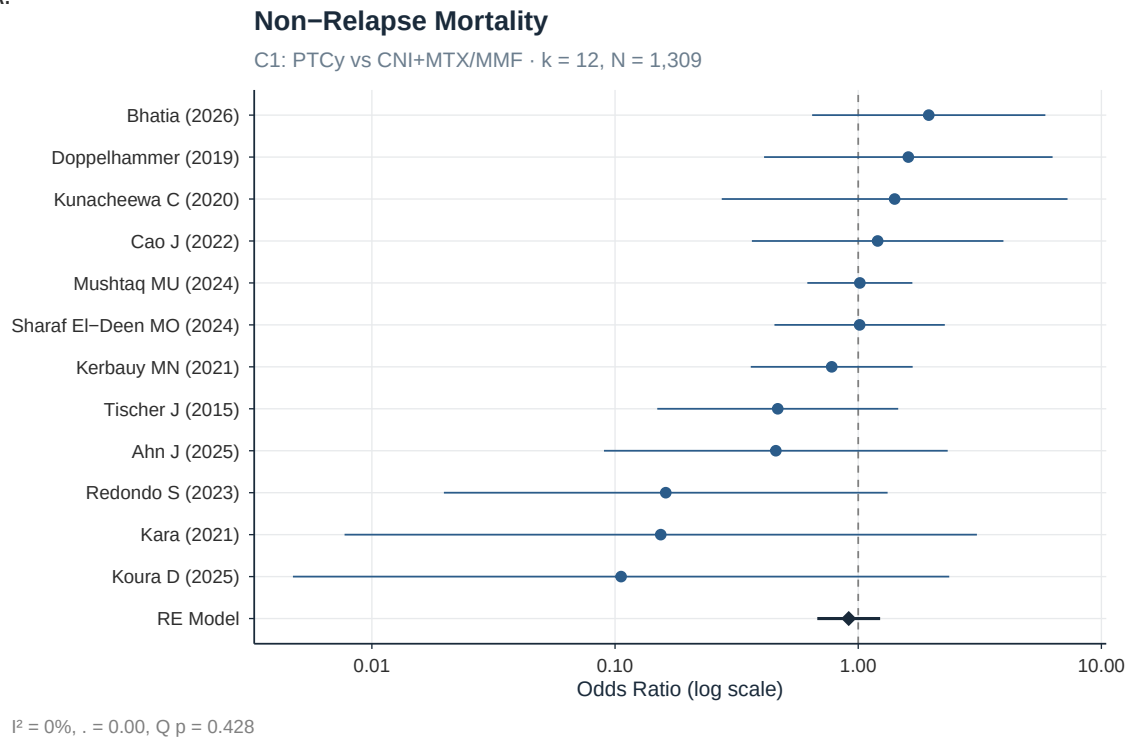


Figure 10: Acute GVHD grade II–IV.

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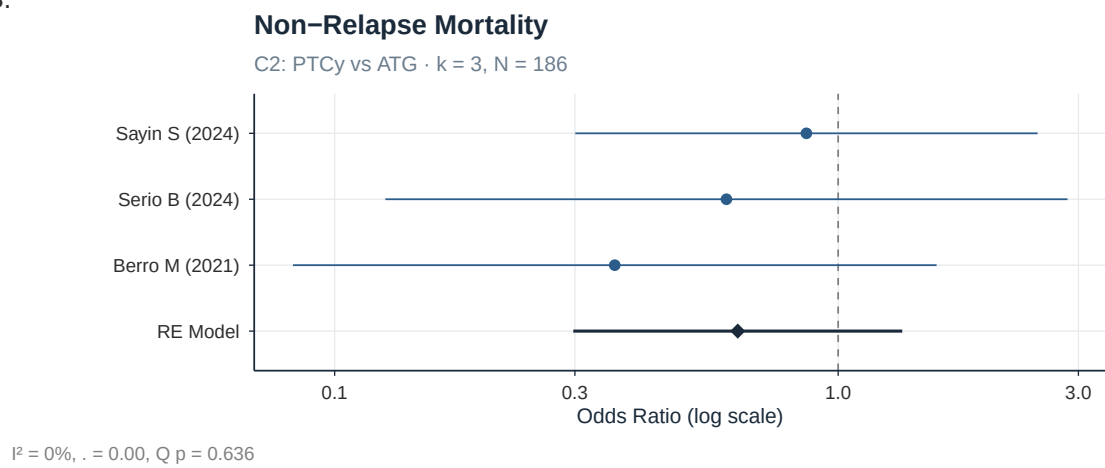
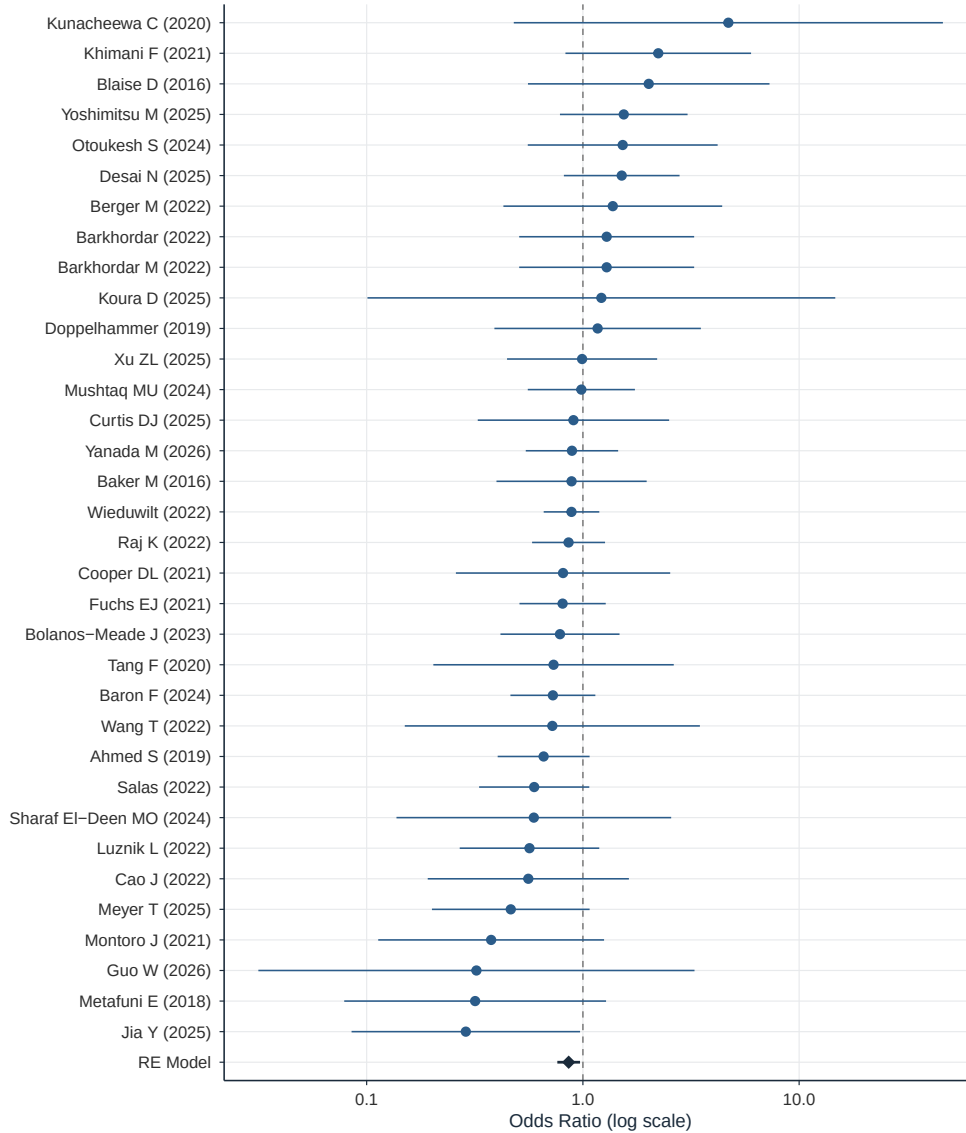


Figure 11: Non-relapse mortality.

A.

Relapse-Related Mortality

C1: PTCy vs CNI+MTX/MMF · k = 34, N = 11,121



B.

Relapse-Related Mortality

C2: PTCy vs ATG · k = 10, N = 17,182

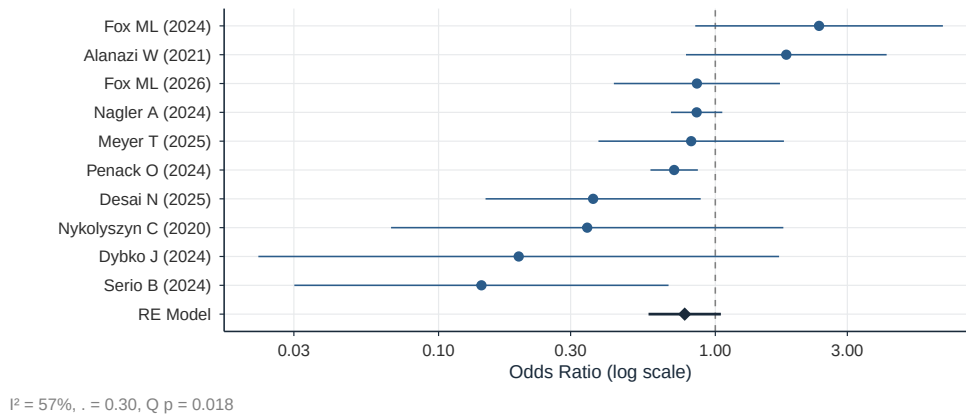
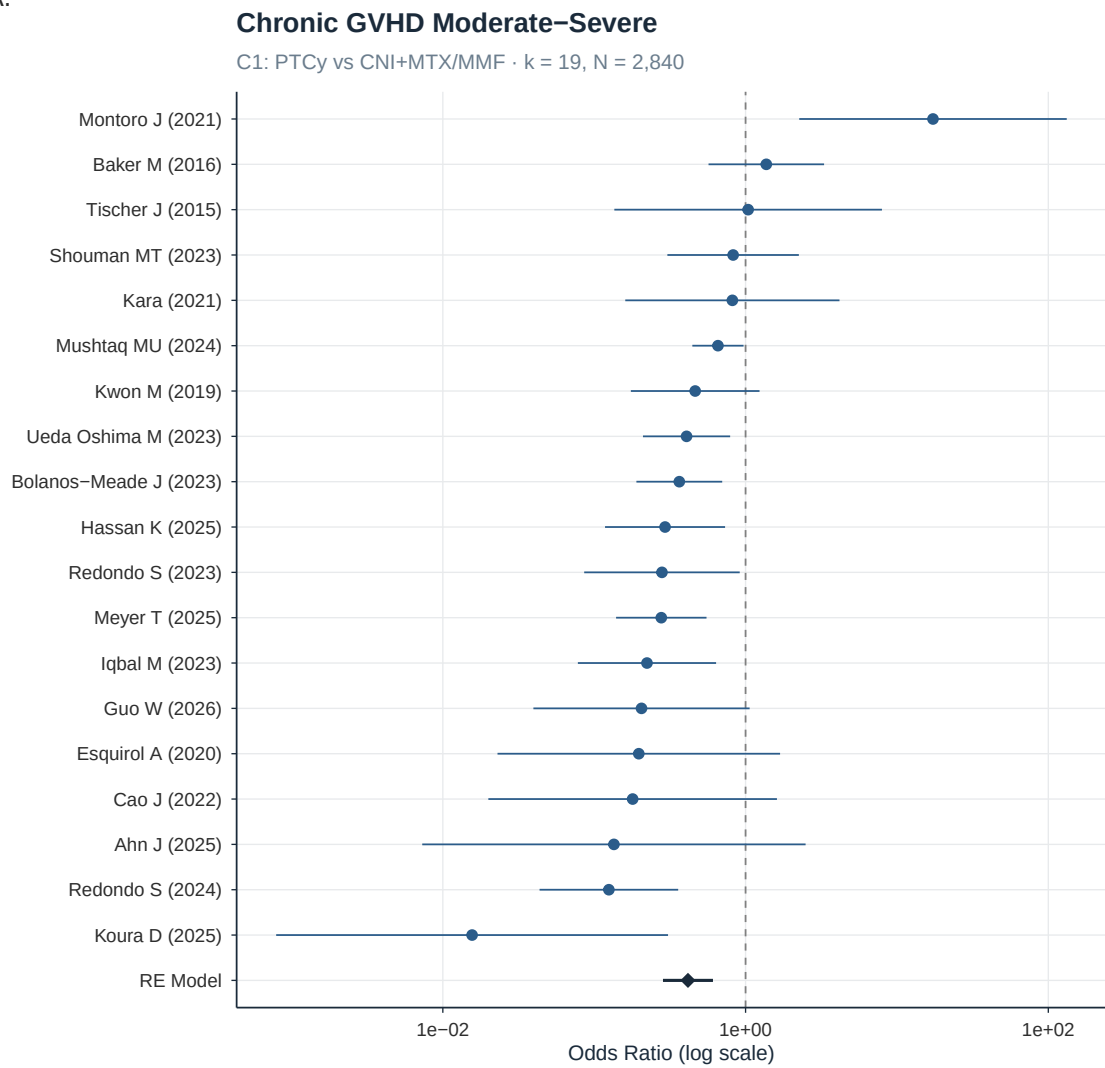


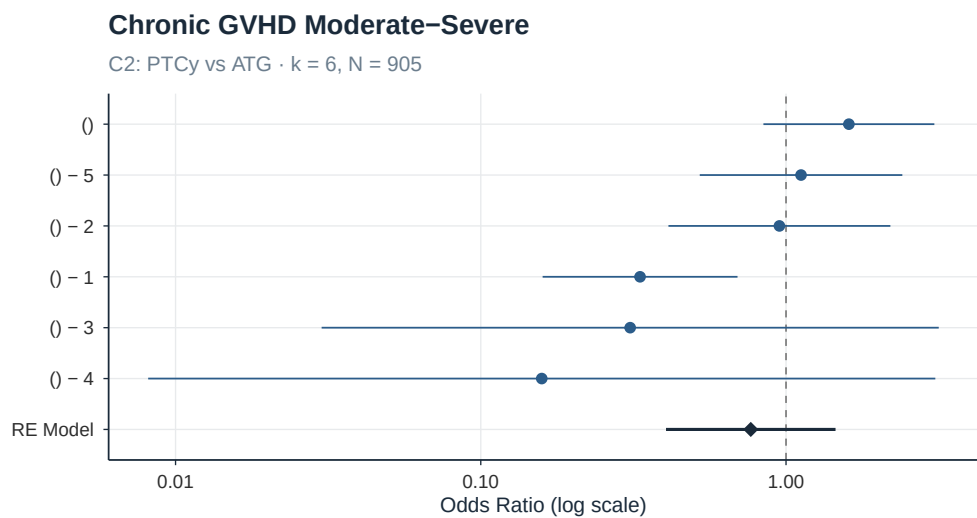
Figure 12: Relapse-related mortality.

A.



$I^2 = 58\%$, $\tau = 0.58$, $Q p < 0.001$

B.



$I^2 = 60\%$, $\tau = 0.57$, $Q p = 0.028$

Figure 13: Chronic GVHD, moderate to severe.

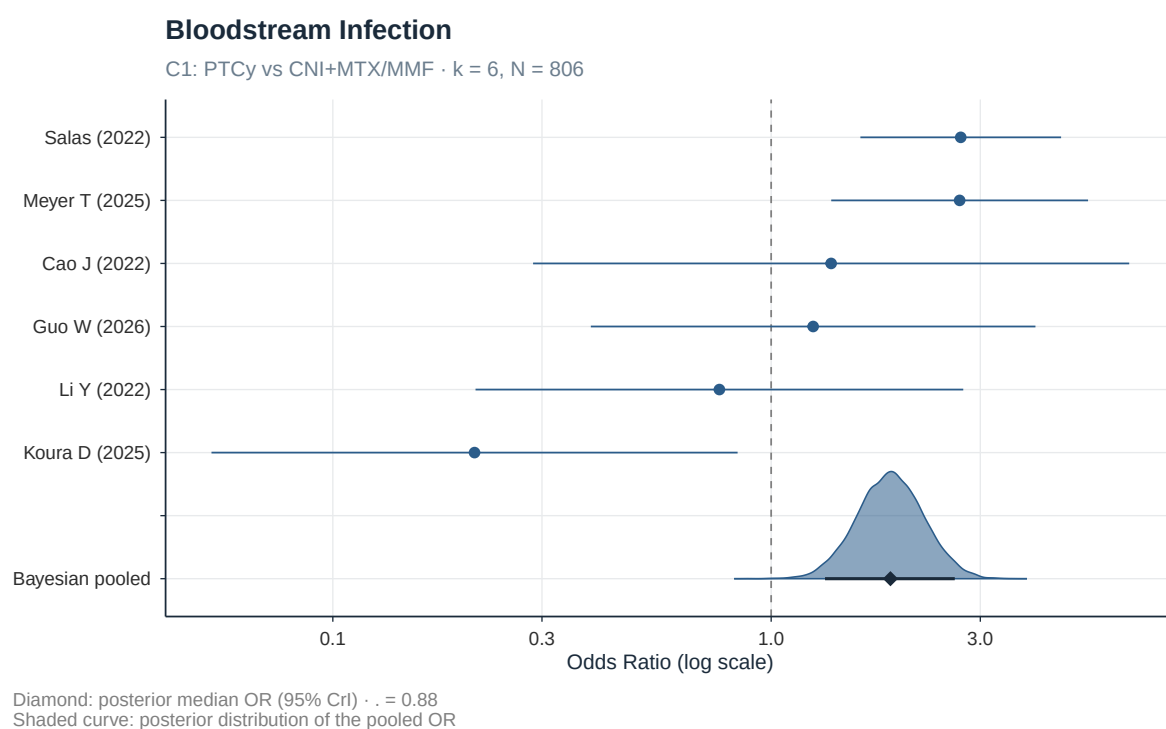


Figure 14: Bloodstream infection, any pathogen.

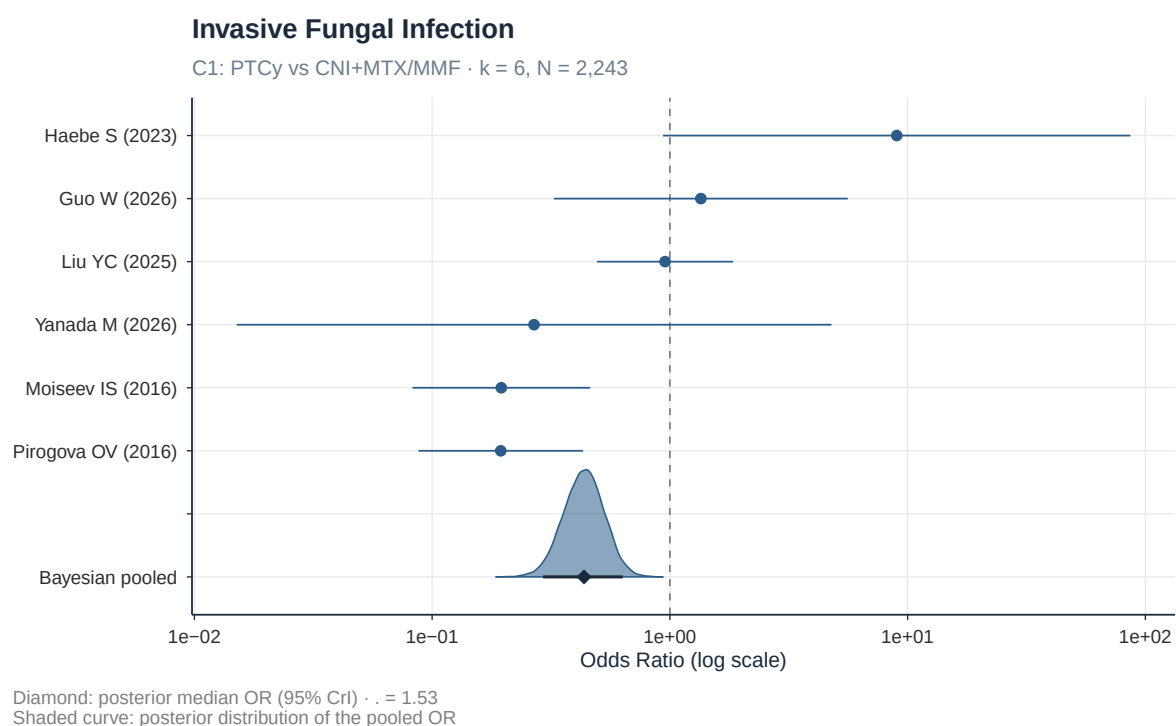
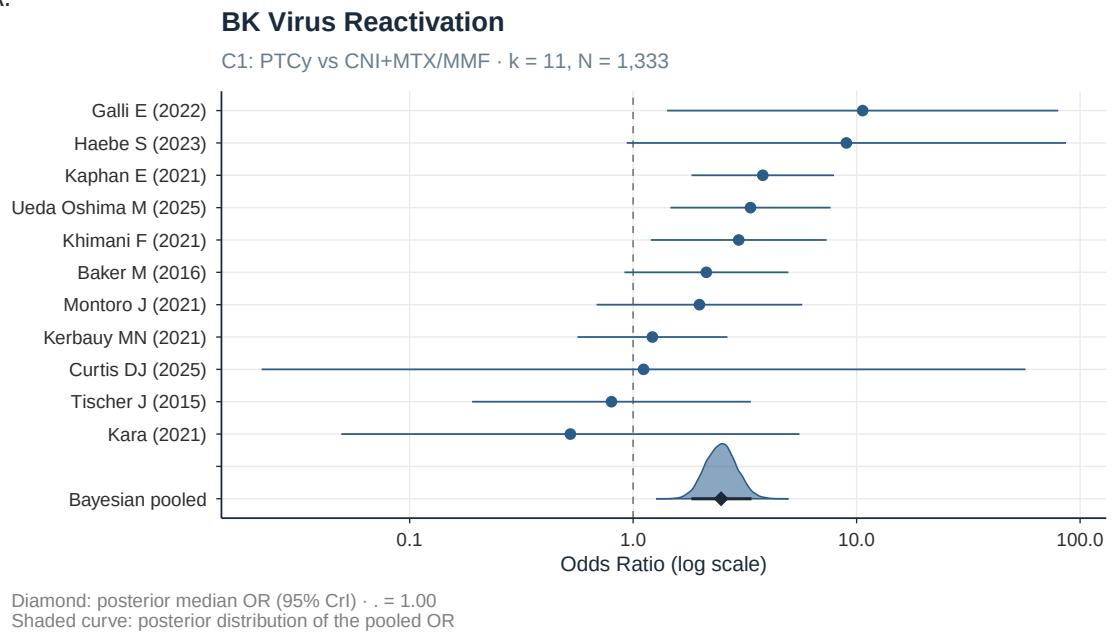


Figure 15: Invasive fungal infection, any.

A.



B.

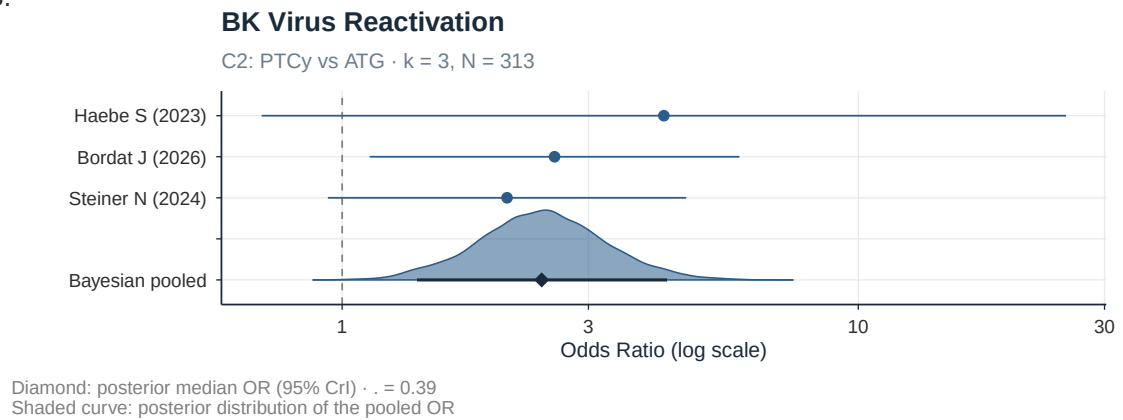
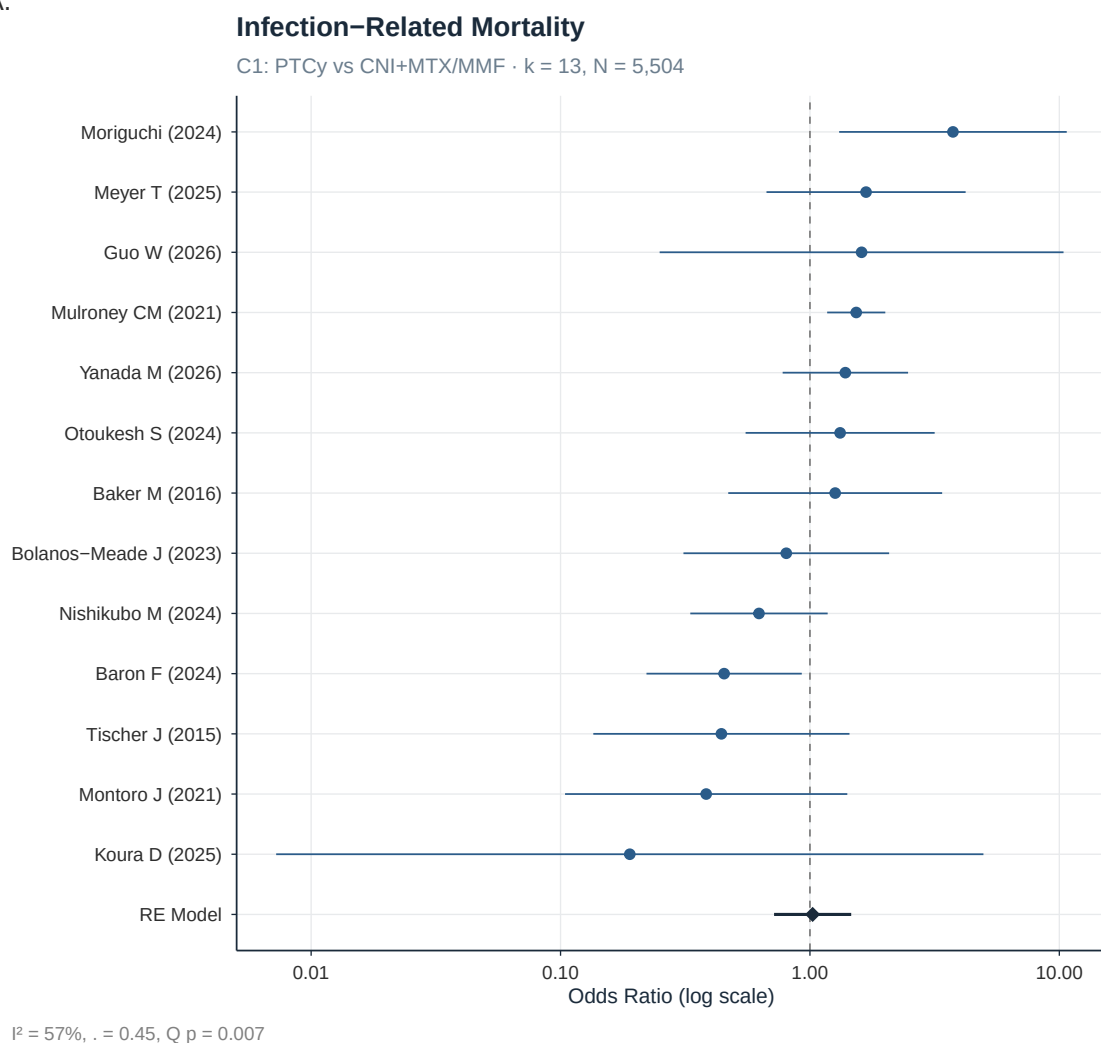


Figure 16: BK virus reactivation.

A.



B.

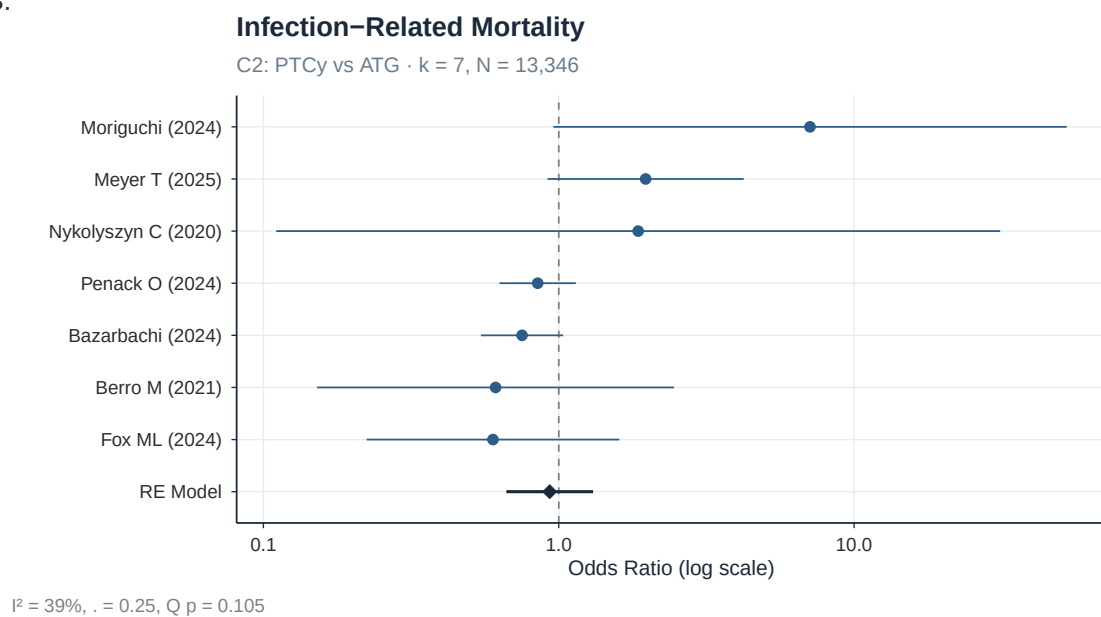


Figure 17: Infection-related mortality.

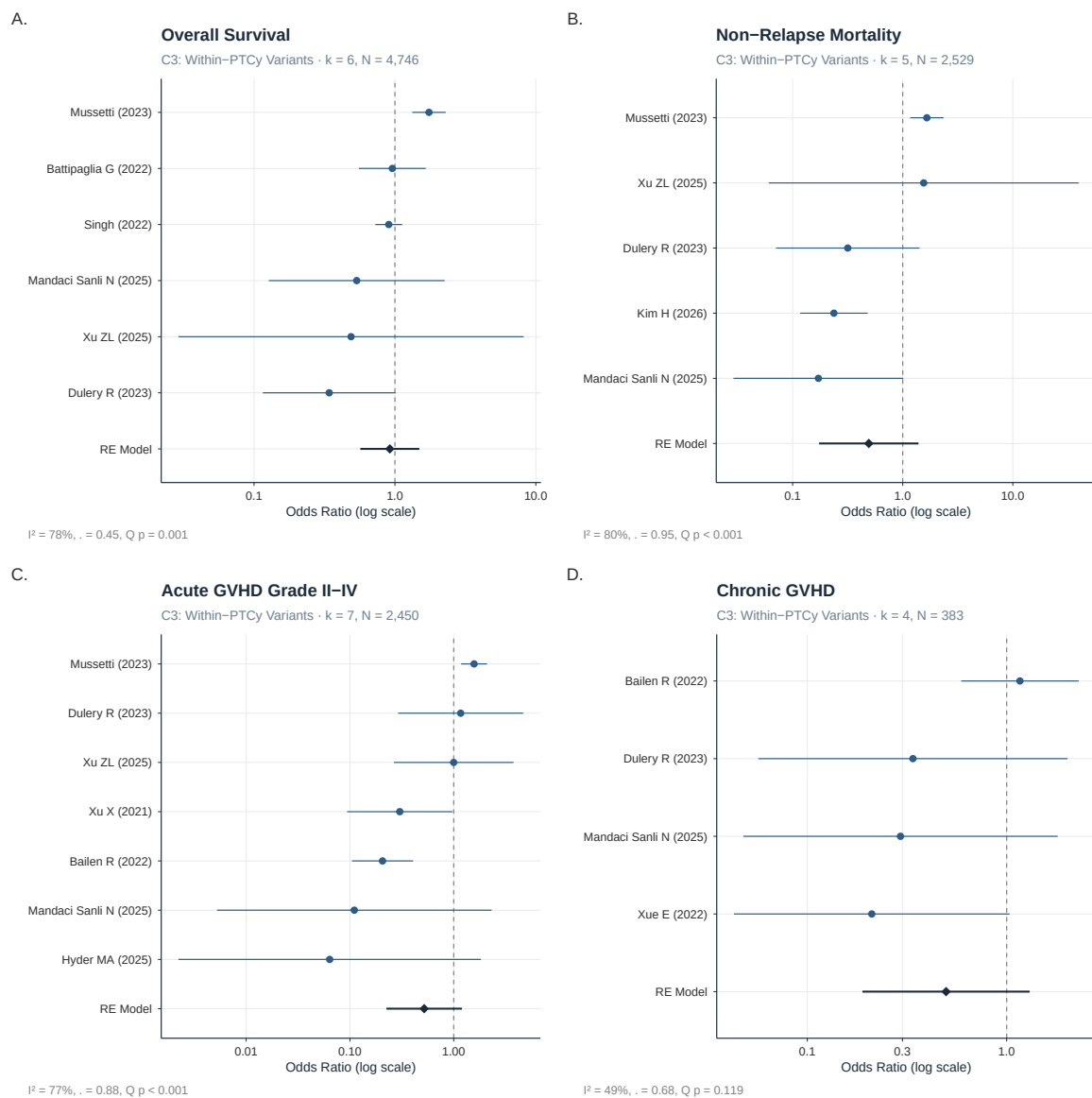


Figure 18: Comparison 3, within-PTCy regimen variants, all outcomes.

117 **S9. Sensitivity analyses**

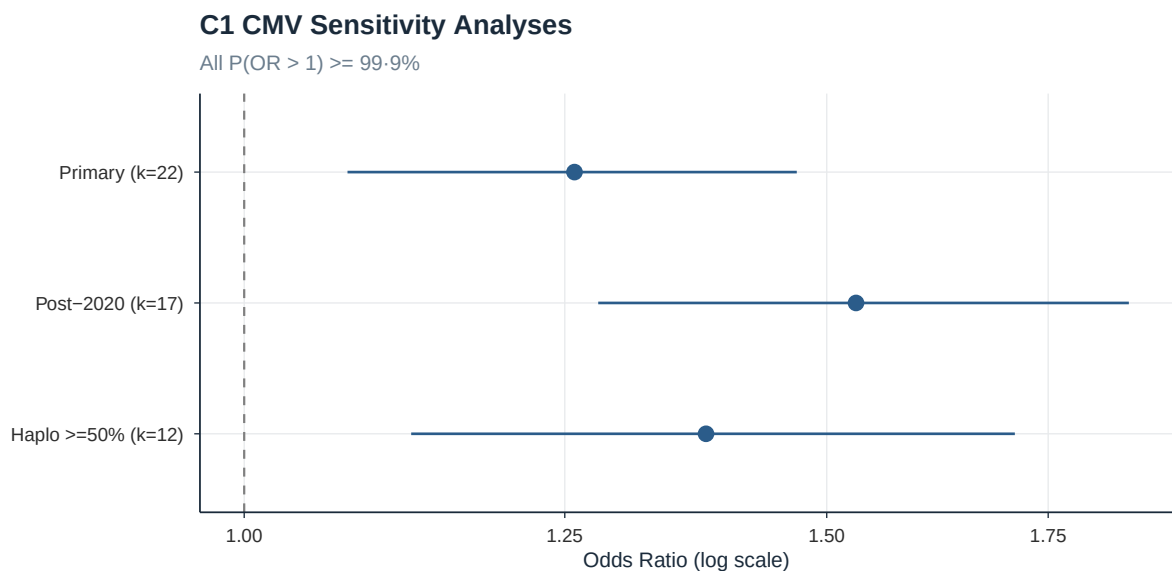


Figure 19: CMV sensitivity models, comparison 1: primary model, post-2020 restriction, and haploid-dominant restriction.

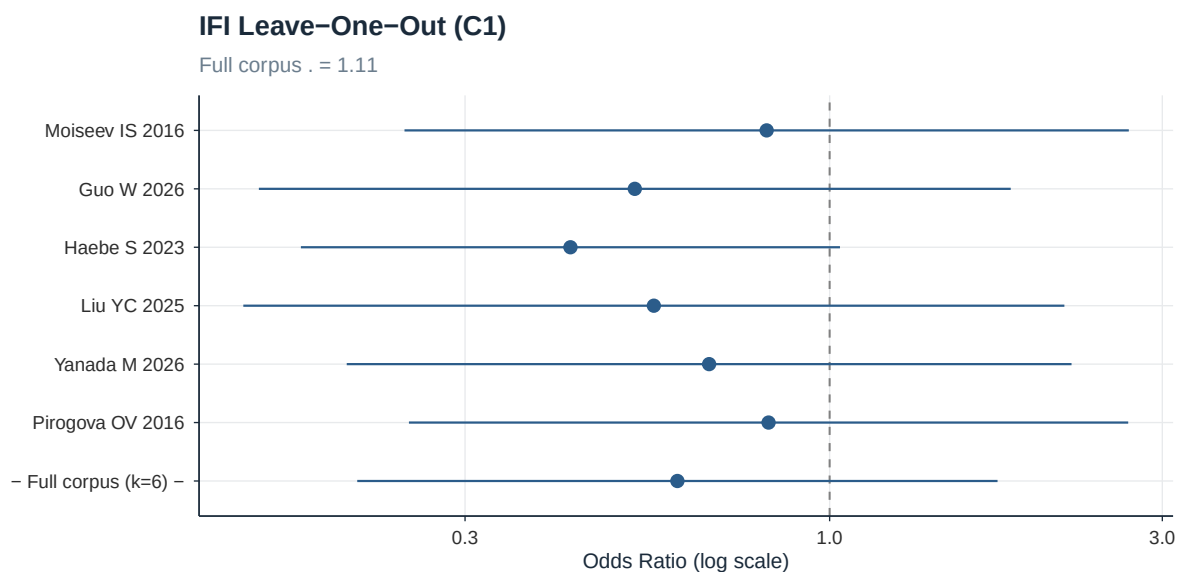


Figure 20: IFI leave-one-out analysis, comparison 1 (frequentist REML).

118 Full frequentist results for all models are provided in `freq_results_setB.csv`, a supple-
 119 mentary data file, which carries the deduplicated fits where deduplication applied and
 120 the original fits otherwise (see its `source` column). Direction was concordant between
 121 Bayesian and frequentist estimates for every outcome. Two framework-specific differ-

ences are worth noting: C1 BSI (Bayesian odds ratio $1 \cdot 87$ versus frequentist $1 \cdot 29$
[$0 \cdot 61$ – $2 \cdot 73$]) and C1 IFI ($0 \cdot 43$ versus $0 \cdot 60$ [$0 \cdot 21$ – $1 \cdot 74$]), both reflecting the influence
of informative priors at small k .

S10. Publication bias

Robust Bayesian model-averaging (RoBMA) models were fitted for all outcomes with k = 6.

S11. GRADE certainty-of-evidence profiles

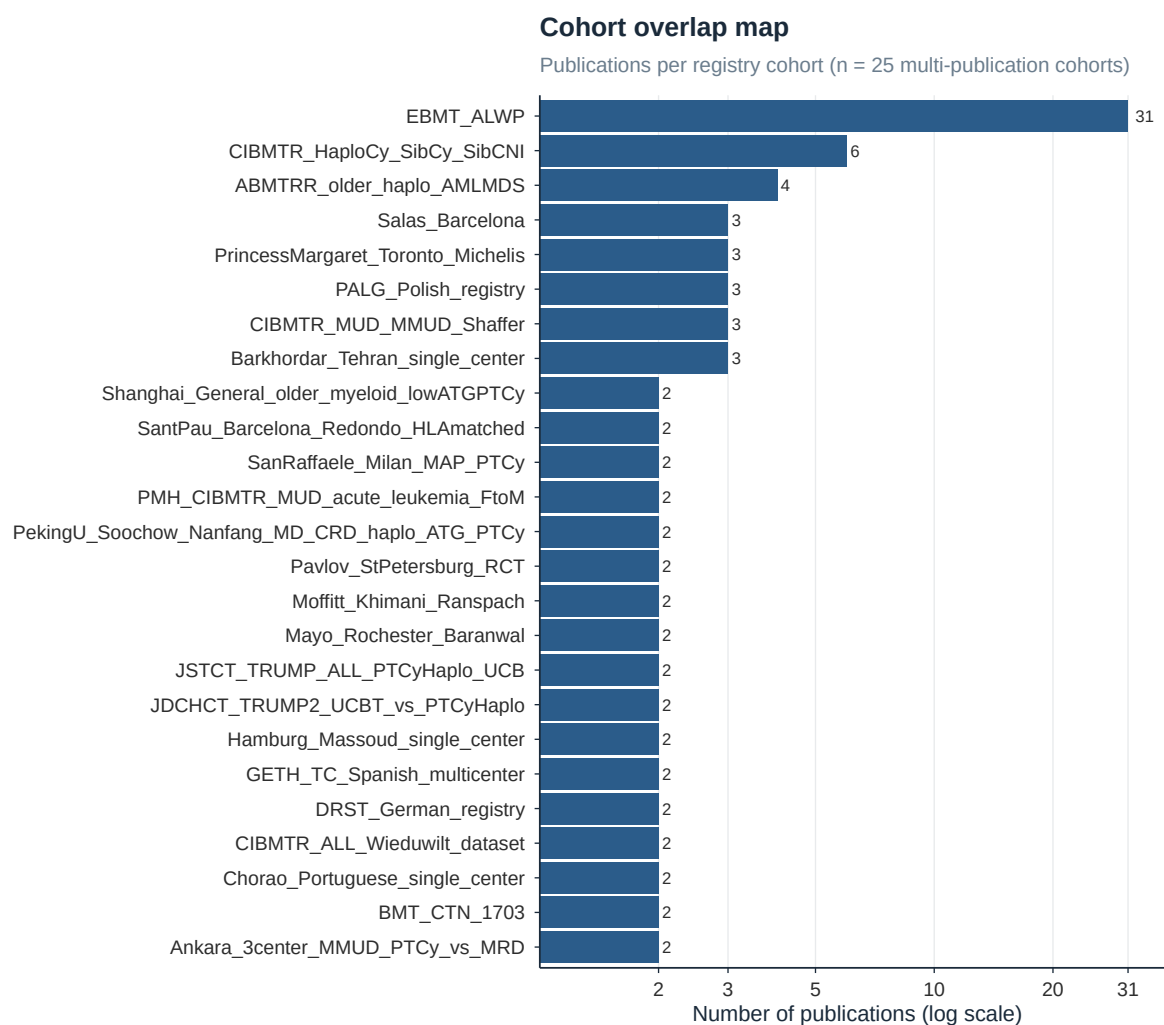


Figure 21: GRADE certainty-of-evidence profiles for the pre-specified primary and secondary outcomes.

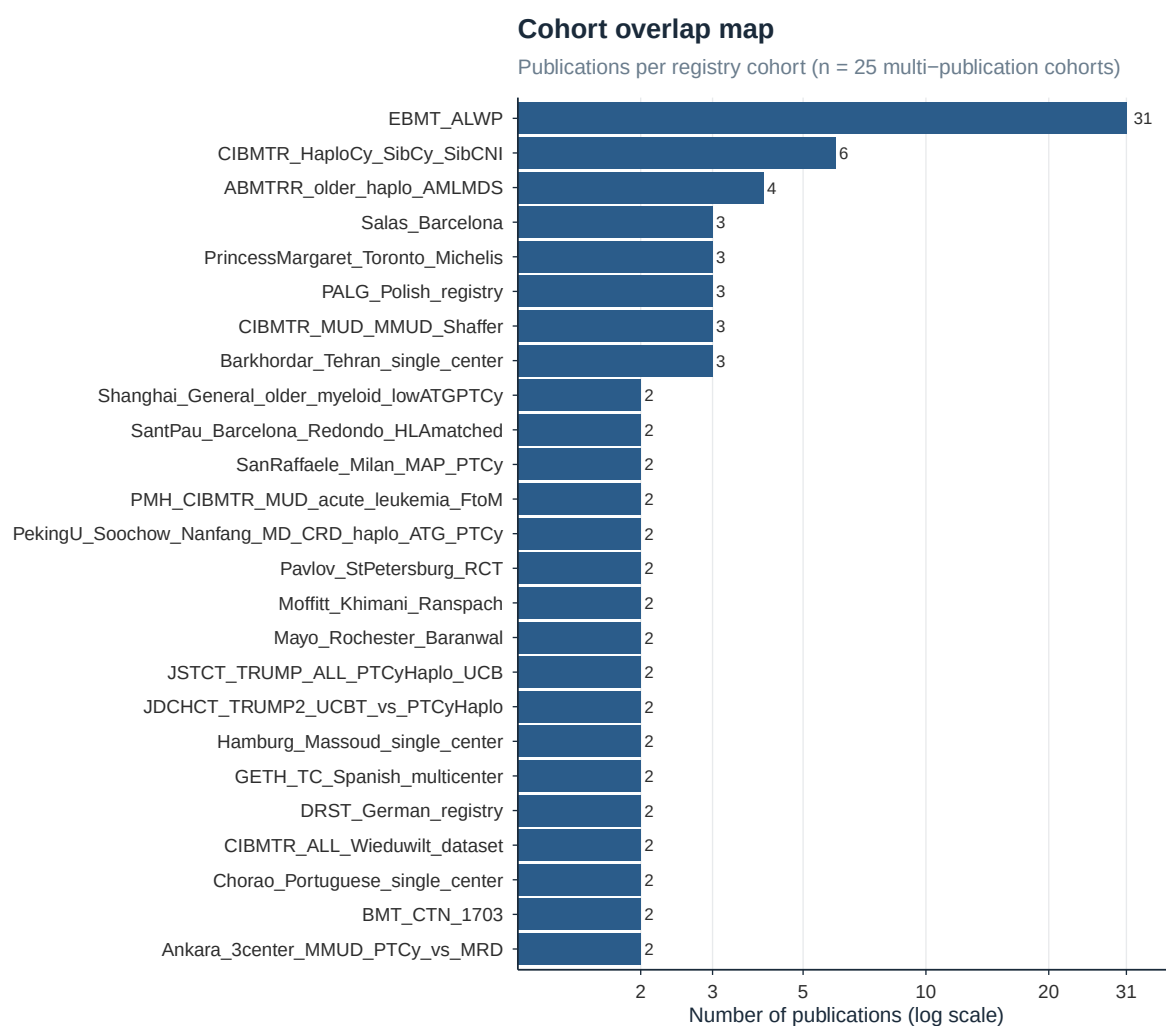


Figure 22: Cohort overlap map. Studies sharing patient cohorts were flagged and the most comprehensive publication designated as primary for each cohort–outcome combination (Set B, the adopted analysis).

129 S12. Cohort overlap

130 S13. PROSPERO registration

131 The study protocol was registered with PROSPERO (CRD420261372156) before data
132 extraction. Protocol amendments were: (1) a Bayesian rather than frequentist primary
133 analysis framework; (2) addition of BK virus reactivation, infection-related mortality,
134 and relapse-related mortality as secondary outcomes; and (3) addition of the steroid-
135 mediation (M2) model.

136 S14. PRISMA 2020 checklist